Homework 5: Reading a Real Option Chain

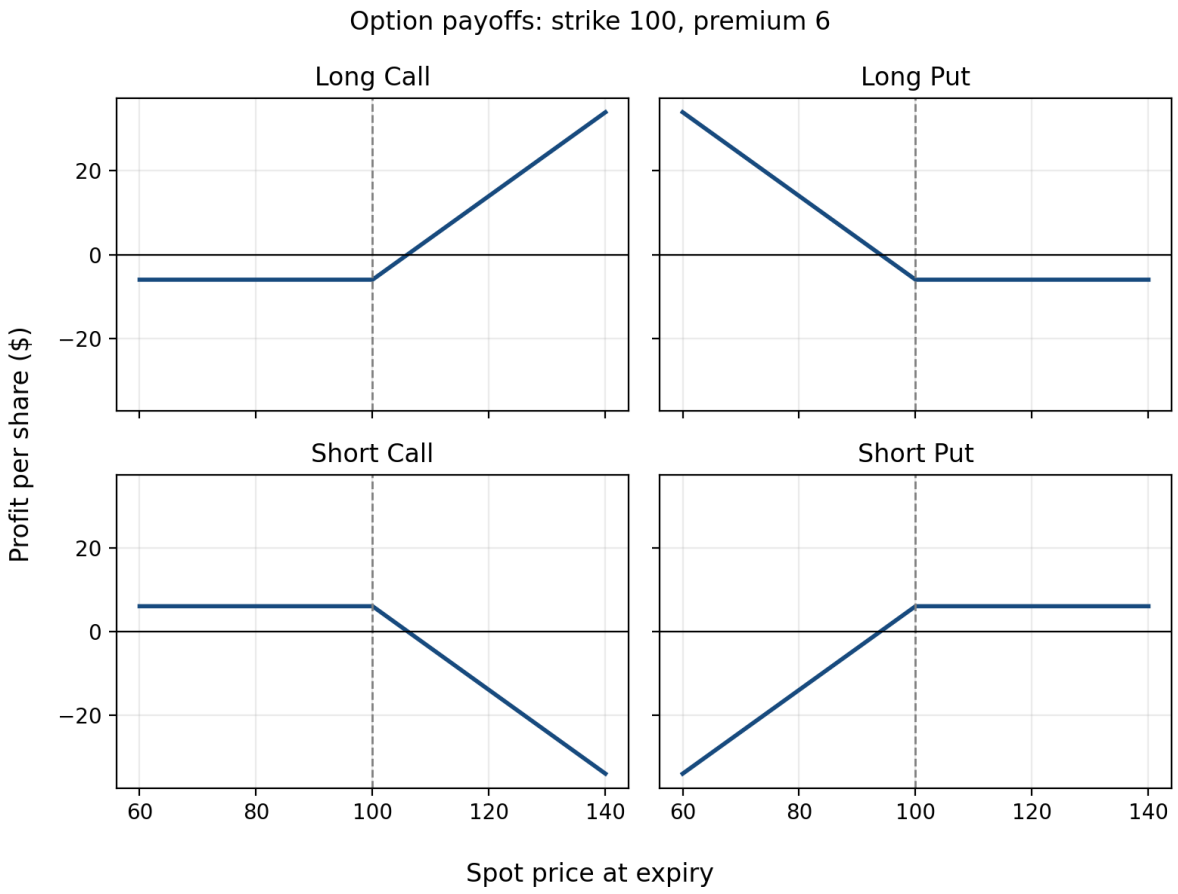
Live-chain snapshot captured 25 October 2026 | SPY latest close: 23 October 2026

1. Payoffs

Position	Spot $80	Spot $100	Spot $120
Long call	-$6.00	-$6.00	+$14.00
Long put	+$14.00	-$6.00	-$6.00
Short call	+$6.00	+$6.00	-$14.00
Short put	-$14.00	+$6.00	+$6.00

All figures are profit per share at expiry for strike $100 and premium $6. Long and short positions are exact mirrors.

2. Payoff diagrams



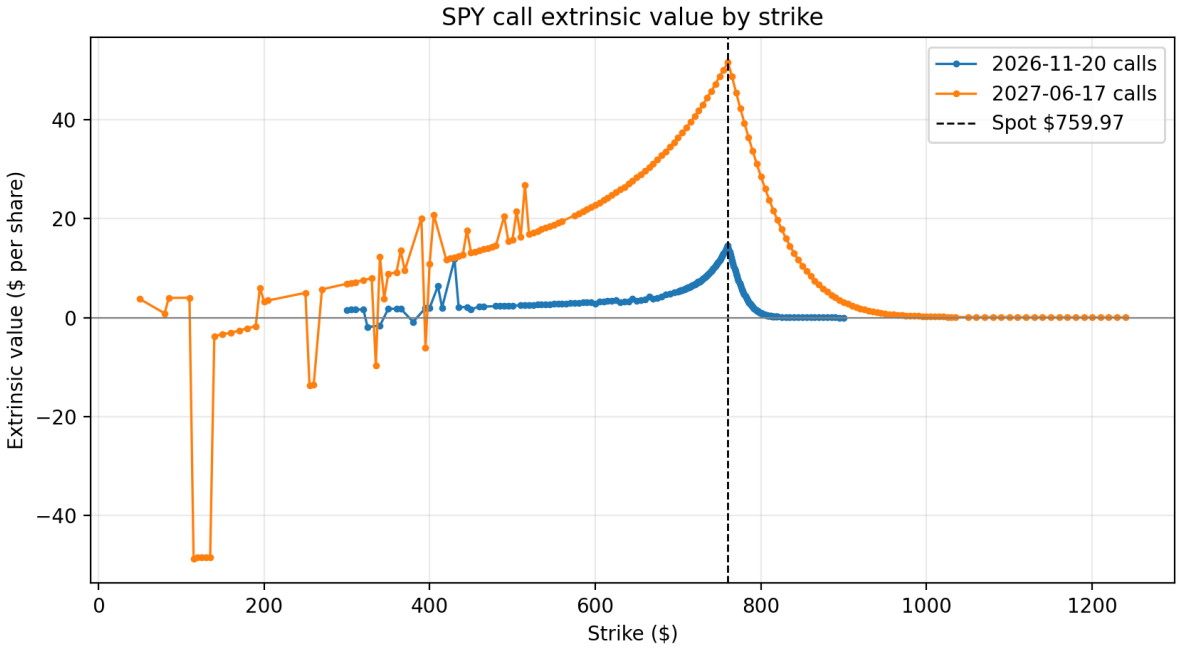
The dashed vertical line is the strike; the horizontal line is zero profit.

3-6. Live SPY call chain

I selected SPY for liquidity. The latest underlying close was **$759.97**. The near expiry was **20 November 2026** (223 calls, 201 puts); the longer expiry was **17 June 2027** (217 calls, 178 puts). Rows with a zero bid were dropped. Mid = (bid + ask) / 2; intrinsic = max(spot - strike, 0); extrinsic = mid - intrinsic.

20 Nov 2026: 10 strikes nearest spot				17 Jun 2027: 10 strikes nearest spot			
Strike	**Mid**	**Intrinsic**	**Extrinsic**	**Strike**	**Mid**	**Intrinsic**	**Extrinsic**
755	17.95	4.97	12.98	735	69.43	24.97	44.46
756	17.26	3.97	13.29	740	65.78	19.97	45.81
757	16.57	2.97	13.60	745	62.20	14.97	47.23
758	15.90	1.97	13.93	750	58.77	9.97	48.80
759	15.24	0.97	14.27	755	55.06	4.97	50.10
760	14.59	0.00	14.59	760	51.75	0.00	51.75
761	13.96	0.00	13.96	765	48.80	0.00	48.80
762	13.34	0.00	13.34	770	45.55	0.00	45.55
763	12.72	0.00	12.72	775	42.41	0.00	42.41
764	12.13	0.00	12.13	780	39.41	0.00	39.41

The largest short-expiry extrinsic value was **$14.59 at $760**; the largest long-expiry extrinsic was **$51.75 at $760**. Both are essentially at the $759.97 spot.



7. Interpretation

Extrinsic value is largest near the current spot price. Those at-the-money calls have little or no exercise value today, but they have the most meaningful chance of ending on either side of the strike by expiry. Far in-the-money calls are already mostly intrinsic value, while far out-of-the-money calls have a smaller chance of becoming valuable, so both tend to carry less time value. In this snapshot, the peaks occur at the $760 strike, almost exactly at SPY spot.

The June curve is much higher because it gives the underlying almost seven extra months to move across the strike. Buying that time means paying for more possible price paths: a currently out-of-the-money call has more opportunity to cross into the money, and an at-the-money call has more opportunity to make a large move. Among 97 common quoted strikes, the long-dated call had higher extrinsic value at 96. The one deep-in-the-money $395 exception displayed negative extrinsic value, an economically implausible artifact of a stale/wide quote rather than a violation of time value.